An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product

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Abstract

We show that the Leech lattice Λ admits the structure of an order in a 24-dimensional real algebra: the lattice is closed under a bilinear product on $\mathbb{R}^{24}$, assembled from nine octonion-multiplication blocks under cyclic $\mathbb{Z}_3$ permutation of three octonion factors. In Robert A. Wilson's sublattice description of Λ, closure turns on a transposition σ of two imaginary basis units: σ leaves the E_8 lattice L invariant and is an octonion-algebra isomorphism, but moves Wilson's sublattices $L\bar{s}$ and Ls. Closure was first observed experimentally, on over 12 million random pairs of minimal vectors; the proof reduces to four lemmas, three of them certified by integer coefficient tables. We determine the idempotents, the automorphism group, and the square-zero elements of the algebra, and the index of the span of the products of the order. We discuss the construction in the context of related work. A historical appendix (1923–1946) traces σ to Bruck's repair of Kirmse's integral octonions. The research record of this human–AI collaboration is public.

Keywords: Leech lattice; integral octonions; E_8 lattice; nonassociative algebra; nonunital order; computer-assisted proof.

MSC 2020: Primary 11H06, 17A75; Secondary 17D05, 68V05.

1 Introduction

The Leech lattice Λ is the unique even unimodular lattice of rank 24 with no vectors of squared norm 2 [ConwaySloane1999]. In the normalization used throughout this paper, with E_8 roots of squared norm 2, Λ is realized as the $\sqrt{2}$-scaled copy of this unimodular lattice. Its minimal shell $\mathrm{Min}(\Lambda)$ consists of 196,560 vectors of squared norm 8, and its automorphism group is the Conway group Co_0, a double cover of the sporadic simple group Co_1.

In rank 8, the E_8 lattice L carries an octonionic multiplication under which it is closed. For a fixed octonion multiplication, exactly seven maximal orders of the rational octonions contain the Gravesian order $\mathbb{Z}\langle e_0, \ldots, e_7\rangle$, each a scaled copy of the E_8 lattice [Coxeter1946, ConwaySmith2003] (Appendix B); the placement of L used here is fixed in Section 2. Wilson [Wilson2009, Section 3] showed that Λ embeds naturally in L^3 via three sublattice conditions, raising the question: does Λ inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Geoffrey M. Dixon [Dixon2010] constructed Λ using an octonionic XY-product. John C. Baez and Greg Egan [Baez2014] explored Jordan-algebraic structures on Λ within the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, finding closure under a doubled Jordan product.

The $\mathbb{Z}_3$-symmetric triple-octonion product on L^3 uses the same octonion multiplication in all three blocks with $\mathbb{Z}_3$-symmetric cross-block routing. It satisfies two of Wilson's three sublattice conditions automatically. The third, however, generally fails: the sublattice it involves is not closed under the standard octonion product, Wilson's included.

The main result of this paper is that closure on Λ is restored by twisting the octonion multiplication on each block by a transposition σ of two imaginary basis units. The map σ fixes the E_8 lattice L setwise and is an octonion-algebra isomorphism, but it moves Wilson's sublattices $L\bar{s}$ and Ls onto sublattices for which the required closure inclusions do hold.

Theorem 1.1. *Let $\star$ denote the $\mathbb{Z}_3$-symmetric triple-octonion product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ under $\mathbb{Z}_3$ cross-block routing (Definition 3.3). Then $u \star v \in \Lambda$ for all $u, v \in \Lambda$ in Wilson's representation [Wilson2009].*

Corollary 1.2. *$(\Lambda, +, \star)$ is an order[1] in the $\mathbb{R}$-algebra $(\mathbb{R}^{24}, +, \star)$.*

Proof. The product $\star$ is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1, $\star$ maps $\Lambda \times \Lambda$ into Λ. □

Remark 1.3 (Order-closure, not shell-closure). The closure statement $\Lambda \star \Lambda \subseteq \Lambda$ means that products land somewhere in Λ; it does not assert that the minimal shell is preserved. The product of two minimal vectors has norm at least 16 (Remark 5.1), never the minimal value $N = 8$: such products land on strictly higher shells, exhaustively for the self-products and on 10^6 sampled distinct pairs. This is consistent with $(\Lambda, +, \star)$ being an order in the ring-theoretic sense: a subring that is a free $\mathbb{Z}$-module of rank equal to the ambient algebra's dimension. That property is weaker than, and independent of, any shell-level closure.

The scope of this paper is deliberately bounded. We establish existence of the order, with a proof that is elementary (four short lemmas, reducing to sublattice inclusions verified in exact arithmetic). Further structural readings are offered as starting points for investigation in Section 5 and Appendix A. Nowhere do we claim or advertise these computer proofs as a substitute for mathematical understanding: they are listed as machine-obtained facts, to provide pointers for the applied reader. Interest in earlier versions of this work came from researchers in medical imaging, cryptography, quantum computing, and phenomenology in the physics of fundamental particles.

All results were developed through a systematic human–AI collaboration, and the complete research record is public [repo]: every prompt, computation, and verification, end to end (Appendix C).

[1] The ambient algebra has no multiplicative identity (Section 5), so the term "order" is used here in the nonunital sense.

2 Preliminaries

2.1 The octonion algebra

Let $\mathbb{O}$ denote the real octonion algebra with basis $\{e_0, e_1, \ldots, e_7\}$, where e_0 is the identity. The multiplication is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{2.1}$$

encoding the rule: for each triple (a,b,c), $e_a \cdot e_b = +e_c$, with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [Baez2002]. Every imaginary unit squares to $-e_0$. Among *unital* real algebras, $\mathbb{O}$ is the unique normed division algebra of dimension 8 (Hurwitz); relaxing "division algebra" to the absence of zero divisors (equivalently, bijectivity of left and right multiplication by every nonzero element) admits further examples, including the nonunital Okubo algebra.

The *norm form* is $N(x) = x\bar{x} = \sum_{i=0}^{7} x_i^2$ for $x = \sum_i x_i e_i$, where $\bar{x} = x_0 e_0 - \sum_{i=1}^{7} x_i e_i$ is the conjugate. Throughout, *norm* means this squared Euclidean length. The composition property $N(xy) = N(x)N(y)$ holds.

2.2 The E_8 lattice

In Wilson's setup [Wilson2009], the E_8 lattice is $L = D_8^+$, the set of vectors in $\mathbb{R}^8$ whose coordinates (in the basis $\{e_0, \ldots, e_7\}$) are either all integers with even coordinate sum, or all half-integers with odd coordinate sum [ConwaySmith2003]. It has 240 roots (vectors of squared norm 2). Under the standard octonion multiplication, L is closed: $L \cdot L \subseteq L$ (Lemma 4.2, certified in Table 1). At this normalization $e_0 \notin L$ (the lattice is even), so L is a nonunital order; as a lattice it is a $\sqrt{2}$-scaled copy of a maximal order of the integral octonions, the unit-scale octavians, for which closure and maximality hold in the classical unital sense [Coxeter1946, ConwaySmith2003]. For a modern survey of the integral-octonion history, see [Petersson2018].

2.3 Wilson's Leech lattice

Define $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$ and $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$, both of norm $N(s) = N(\bar{s}) = 2$. Then $s \in L$ (it is a root in the half-integer branch, with an odd coordinate sum), while $\bar{s} \notin L$ (its coordinate sum is -4, which falls in the excluded even branch of the half-integer case). Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \qquad Ls = \{\ell \cdot s : \ell \in L\}. \tag{2.2}$$

Both $L\bar{s}$ and Ls are sublattices of L,[2] with $[L : L\bar{s}] = [L : Ls] = 16$, $L\bar{s} + Ls = L$, and $L\bar{s} \cap Ls = 2L$.

Definition 2.1 (Wilson)**.** The *Leech lattice* is

$$\Lambda = \big\{(x,y,z) \in L^3 : (1)\ x,y,z \in L;\ (2)\ x{+}y,\ x{+}z,\ y{+}z \in L\bar{s};\ (3)\ x{+}y{+}z \in Ls\big\}. \tag{2.3}$$

[2] $L\bar{s} \subseteq L$ even though $\bar{s} \notin L$: each $\ell \cdot \bar{s}$ with ℓ a root of L lies in L by direct computation, and the inclusion extends by $\mathbb{Z}$-linearity in the left factor. $L\bar{s}$ is therefore a $\mathbb{Z}$-subgroup of L, and is moreover closed under left-multiplication by L (Wilson's condition (2) for the untwisted product, implicit in the sublattice argument). For Ls, the subgroup property is likewise immediate from $s \in L$. Unlike $L\bar{s}$, the sublattice Ls is not closed under the octonion product itself ($Ls \cdot Ls \not\subseteq Ls$). That failure is precisely the obstruction resolved in this paper (Remark 4.5).

The 196,560 minimal vectors of Λ decompose into three families:

Family	Formula	Count
Type 1	$(2\lambda, 0, 0)$ and cyclic permutations	$3 \times 240 = 720$
Type 2	$(\lambda\bar{s},\ (\lambda\bar{s})j,\ 0)$ and cyclic permutations	$3 \times 240 \times 16 = 11{,}520$
Type 3	$((\lambda s)j,\ \lambda k,\ (\lambda j)k)$ and cyclic permutations	$3 \times 240 \times 16^2 = 184{,}320$

Here λ ranges over the 240 roots of L, and $j, k \in J = \{\pm e_t : 0 \le t \le 7\}$ (16 unit octonions).

3 The construction

Definition 3.1 (Transposition)**.** Let $\sigma = (i\ j)$ be a transposition on $\{1, \dots, 7\}$, extended to the linear involution $\sigma\colon \mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and swaps $e_i \leftrightarrow e_j$.

Definition 3.2 (σ-twisted octonion product)**.** The *σ-twisted octonion product* $\cdot_\sigma$ on $\mathbb{O}$ is the σ-twisted variant of the standard octonion product (2.1):

$$x \cdot_\sigma y \ :=\ \sigma\big(\sigma(x) \cdot \sigma(y)\big). \tag{3.1}$$

This is another octonion product on $\mathbb{O}$: by definition, $\sigma\colon (\mathbb{O}, \cdot) \to (\mathbb{O}, \cdot_\sigma)$ is an algebra isomorphism, $\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y)$.

Definition 3.3 (Triple product with $\mathbb{Z}_3$ routing)**.** Decompose $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ into three copies of $\mathbb{R}^8$ (blocks indexed $\alpha = 1, 2, 3$). For $u = (x, y, z)$ and $v = (x', y', z')$ with $x, x' \in \mathbb{O}_1$, etc., define the *triple product*

$$u \star v = (P_1,\ P_2,\ P_3), \tag{3.2}$$

where each block P_α is the sum of three σ-twisted octonion products:

$$P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z', \tag{3.3}$$
$$P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x', \tag{3.4}$$
$$P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'. \tag{3.5}$$

The routing rule is: a product of blocks α and β lands in block γ, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ for $\alpha \neq \beta$, and in block α when $\alpha = \beta$. All nine block-pair products use the same multiplication $\cdot_\sigma$.

4 Proof of closure

We prove Theorem 1.1 by showing that the σ-twisted octonion product preserves each of Wilson's three conditions (Definition 2.1). Throughout, and without loss of generality, $\sigma = (1\ 2)$.[3] From the given choice of octonion basis and Fano triples, the load-bearing effect of σ is that it *moves* the sublattices $L\bar{s}$ and Ls, while fixing L setwise (Lemma 4.1) and carrying the octonion product to an isomorphic one (Definition 3.2). Lemmas 4.3 and 4.4, the closure inclusions for the moved sublattices, are where the work of the proof lives.

[3] The 21 transpositions of imaginary basis units form a single orbit under conjugation by the order-21 group of coordinate permutations preserving the oriented Fano triples (2.1). Each such permutation is an octonion automorphism that fixes s and $\bar{s}$ and preserves L, so conjugating by it carries the proof below for (1 2) to any other transposition.

4.1 The four lemmas

Lemma 4.1. $\sigma(L) = L$.

Proof. The lattice $L = D_8^+$ consists of all vectors in $\mathbb{R}^8$ whose coordinates are either all integers with even coordinate sum, or all half-integers with odd coordinate sum (Section 2). The transposition σ permutes two coordinates, which preserves both the integer/half-integer parity and the coordinate sum. Hence $\sigma(L) \subseteq L$. Since σ is an involution, $\sigma(L) = L$. □

For each of the next three lemmas, L, $\sigma(L\bar{s})$, and $\sigma(Ls)$ are full-rank $\mathbb{Z}$-lattices of rank 8 in $\mathbb{R}^8$. By $\mathbb{Z}$-bilinearity of the octonion product, an inclusion $A \cdot B \subseteq C$ between such lattices holds if and only if it holds on a chosen pair of $\mathbb{Z}$-bases: if $\{a_i\}_{i=1}^8$ generates A and $\{b_j\}_{j=1}^8$ generates B, then $A \cdot B \subseteq C$ is equivalent to each of the 64 products $a_i \cdot b_j$ lying in $\mathbb{Z}\langle c_1, \ldots, c_8\rangle$ for a chosen $\mathbb{Z}$-basis $\{c_k\}_{k=1}^8$ of C. (The forward direction is the substantive one and uses bilinearity; the reverse direction is immediate since each $a_i \in A$ and each $b_j \in B$.) Each lemma below is proved by exhibiting that finite list of integer coefficients explicitly. Concrete $\mathbb{Z}$-bases for L, $\sigma(Ls)$, and $\sigma(L\bar{s})$ are fixed in Appendix A (denoted there L_k, M_k, N_k respectively); the 64-entry coefficient tables for the three lemmas appear as Tables 1, 2, and 3.

Lemma 4.2. $L \cdot L \subseteq L$.

Proof. The 64 products $L_i \cdot L_j$ are integer combinations of $\{L_1, \ldots, L_8\}$; the explicit coefficients are Table 1. By $\mathbb{Z}$-bilinearity, $L \cdot L \subseteq L$. This is the norm-2-scale form of the classical fact that the maximal orders of the integral octonions are closed under multiplication [Coxeter1946, ConwaySmith2003]. □

Lemma 4.3. $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$.

Proof. The 64 products $L_i \cdot N_j$ are integer combinations of $\{N_1, \ldots, N_8\}$; the explicit coefficients are Table 2. By $\mathbb{Z}$-bilinearity, $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$. □

Lemma 4.4. $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$.

Proof. The 64 products $M_i \cdot M_j$ are integer combinations of $\{M_1, \ldots, M_8\}$; the explicit coefficients are Table 3. By $\mathbb{Z}$-bilinearity, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$. □

Remark 4.5. The sublattice Ls is *not* closed under the standard octonion product: $u_1 = (-1,0,0,0,1,0,1,1)$ and $u_2 = (-1,1,0,0,0,1,0,1)$ lie in Ls; $u_1 \cdot u_2 = (0,-2,2,1,-2,-1,-1,-1)$ lies in $L\bar{s} \setminus Ls$. Moreover $\sigma(Ls) \neq Ls$: with $\sigma = (1\ 2)$, the element $v = (-1,0,1,0,0,1,0,1)$ lies in $\sigma(Ls) \setminus Ls$. The twist therefore moves the block-sum sublattice off Ls onto the closed sublattice $\sigma(Ls)$ (Lemma 4.4). Likewise $\sigma(L\bar{s}) \neq L\bar{s}$: $w = (0,0,-1,0,-1,0,-1,-1) \in \sigma(L\bar{s}) \setminus L\bar{s}$.

4.2 Blockwise form of the triple product

Since each σ-twisted product has the form $a \cdot_\sigma b = \sigma(\sigma(a) \cdot \sigma(b))$ and σ is linear, each block P_α is σ applied to a sum of standard octonion products of σ-images of the inputs. Write $X = \sigma(x)$, $Y = \sigma(y)$, $Z = \sigma(z)$, $X' = \sigma(x')$, $Y' = \sigma(y')$, $Z' = \sigma(z')$:

$$P_1 = \sigma\big(X \cdot X' + Z \cdot Y' + Y \cdot Z'\big), \tag{4.1}$$

$$P_2 = \sigma\big(Y \cdot Y' + X \cdot Z' + Z \cdot X'\big), \tag{4.2}$$

$$P_3 = \sigma\big(Z \cdot Z' + Y \cdot X' + X \cdot Y'\big). \tag{4.3}$$

Taking sums:

- **Block sum.** $P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big)$.
- **Pairwise sums.** For example, $P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big)$, and similarly for the other two pairs.

4.3 Wilson's condition (1): each block lies in L

Proposition 4.6. *For all $u, v \in \Lambda$, each block P_α of $u \star v$ lies in L.*

Proof. Consider $P_1 = \sigma(X \cdot X' + Z \cdot Y' + Y \cdot Z')$. Since $u \in \Lambda$, condition (1) gives $x \in L$, so $X = \sigma(x) \in L$ by Lemma 4.1. Similarly $X', Y, Y', Z, Z' \in L$. By Lemma 4.2, each product $X \cdot X'$, $Z \cdot Y'$, $Y \cdot Z'$ lies in L. Their sum lies in L (closed under addition). Applying σ and using Lemma 4.1, $P_1 \in L$. The same argument applies to P_2 and P_3. □

4.4 Condition (2): pairwise sums lie in $L\bar{s}$

Proposition 4.7. *For all $u, v \in \Lambda$, the pairwise sums $P_\alpha + P_\beta$ lie in $L\bar{s}$.*

Proof. We show $P_1 + P_2 \in L\bar{s}$; the other two cases follow by the $\mathbb{Z}_3$ symmetry of the routing.

From (4.1)–(4.2):

$$P_1 + P_2 = \sigma\big(X \cdot (X' + Z') + Y \cdot (Y' + Z') + Z \cdot (X' + Y')\big). \tag{4.4}$$

Since $v \in \Lambda$, condition (2) gives $x' + z' \in L\bar{s}$, so $X' + Z' = \sigma(x' + z') \in \sigma(L\bar{s})$. Similarly $Y' + Z' \in \sigma(L\bar{s})$ and $X' + Y' \in \sigma(L\bar{s})$. Since $u \in \Lambda$, we have $X, Y, Z \in L$ (by Lemma 4.1).

By Lemma 4.3, each product $X \cdot (X' + Z')$, $Y \cdot (Y' + Z')$, $Z \cdot (X' + Y')$ lies in $\sigma(L\bar{s})$. Their sum lies in $\sigma(L\bar{s})$. Applying the outer σ:

$$P_1 + P_2 = \sigma(\text{element of } \sigma(L\bar{s})) \in \sigma(\sigma(L\bar{s})) = L\bar{s}, \tag{4.5}$$

since $\sigma^2 = \mathrm{id}$. □

4.5 Condition (3): the block sum lies in Ls

This is the crux of the proof. For the standard (untwisted) product the block sum would require $Ls \cdot Ls \subseteq Ls$, which does not hold (Remark 4.5). The twist redirects the computation through $\sigma(Ls)$, where closure holds.

Proposition 4.8. *For all $u, v \in \Lambda$, the block sum $P_1 + P_2 + P_3$ lies in Ls.*

Proof. From (4.1)–(4.3):

$$P_1 + P_2 + P_3 = \sigma\big((X + Y + Z) \cdot (X' + Y' + Z')\big). \tag{4.6}$$

Since $u \in \Lambda$, condition (3) gives $x + y + z \in Ls$, so

$$X + Y + Z = \sigma(x + y + z) \in \sigma(Ls). \tag{4.7}$$

Similarly $X' + Y' + Z' \in \sigma(Ls)$.

By Lemma 4.4, $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$, so

$$(X + Y + Z) \cdot (X' + Y' + Z') \in \sigma(Ls). \tag{4.8}$$

Applying the outer σ:

$$P_1 + P_2 + P_3 \in \sigma(\sigma(Ls)) = Ls. \tag{4.9}$$

□

This completes the proof of Theorem 1.1.

5 Algebraic properties

Throughout this section, $\sigma = (1\ 2)$, as in Section 4.

5.1 Identities

Several standard algebraic identities, among them associativity, alternativity, flexibility, power-associativity and commutativity, are easily checked to fail on the order $(\Lambda, +, \star)$. To give an initial idea of the degree of their violation, we have drawn samples uniformly (by family weight) from $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ for the binary identities, from $\mathrm{Min}(\Lambda)$ for the cube and quartic identities, and from $\mathrm{Min}(\Lambda)^3$ for the symmetric-composition identity. Sample size is $N = 10^6$ per property; exact integer arithmetic throughout.[4] The standard error is $\mathrm{SE} = \sqrt{p(1-p)/N}$, where p is the fraction of samples on which the identity holds.

Property	Holds?	Fraction of samples (% ± SE)
Multiplicative identity	No	n/a (structural)
Commutativity $u \star v = v \star u$	No	$(0.018 \pm 0.0014)\%$
Norm multiplicativity $N(u \star v) = N(u)N(v)$	No	$(46.77 \pm 0.050)\%$
Left alternativity $(x \star x) \star y = x \star (x \star y)$	No	$(0.188 \pm 0.0043)\%$
Right alternativity $(x \star y) \star y = x \star (y \star y)$	No	$(0.188 \pm 0.0043)\%$
Flexibility $(x \star y) \star x = x \star (y \star x)$	No	$(8.37 \pm 0.028)\%$
Cube power-associativity $x^2 \star x = x \star x^2$	No	$(30.70 \pm 0.046)\%$
Quartic power-associativity[5]	No	$(12.75 \pm 0.033)\%$
Symmetric composition $\langle u \star v, w\rangle = \langle u, v \star w\rangle$	No	$(25.37 \pm 0.044)\%$

The failure of norm multiplicativity rules out $(\Lambda, +, \star)$ being a composition algebra; the failure of symmetric composition rules out, independently, its being a symmetric composition algebra in the sense of Okubo and para-Hurwitz (Alberto Elduque [Elduque1996, Elduque2023]; Susumu Okubo [Okubo1995]). Cube and quartic power-associativity are reported separately; the quartic identity was strictly stronger on every sample: in the joint exact-arithmetic count over the same 10^6 samples, no quartic-passing sample failed the cube identity, while the cube-passing set was roughly $2.4\times$ larger.

[4] [repo] `python_project/src/verify_section5_properties.py`.
[5] All five degree-4 parenthesizations of $x \star x \star x \star x$ coincide.

Remark 5.1. Exhaustively over the minimal shell, the self-product norms $N(u \star u)$ of the 196,560 minimal vectors take values in $\{16, 32, 48, 64, 80, 96, 128\}$: neither 0 nor 8 occurs, so the minimal shell has no idempotent and no square-zero vector.[6] On 10^6 sampled pairs, the pair-product norms $N(u \star v)$ took only the values $\{16, 32, 48, 64, 80, 96, 112, 128\}$, all multiples of 16, with mode 64 (the norm-multiplicative value $N(u)N(v) = 8 \times 8$).[7] Both shell questions are settled on all of Λ in Sections 5.2 and 5.4: the order has no nonzero idempotent; it does, however, contain square-zero vectors, the smallest at norm 12.

Remark 5.2. The ambient algebra $(\mathbb{R}^{24}, +, \star)$ has no multiplicative identity: any putative left identity (a, b, c) would have to act on an arbitrary (x', y', z') by the block matrix $\left(\begin{smallmatrix} a & c & b \\ c & b & a \\ b & a & c \end{smallmatrix}\right)$ (entries acting by left-multiplication in $(\mathbb{O}, \cdot_\sigma)$), and no triple (a, b, c) makes this the identity on all three blocks simultaneously. In particular, (e_0, e_0, e_0) is not an identity: it maps (x', y', z') to $(x' + y' + z',\ x' + y' + z',\ x' + y' + z')$. Consequently $(\Lambda, +, \star)$ is a nonunital order.

The defining property of the order is *closure*: the Leech lattice is stable under the product.

5.2 Idempotents

The ambient algebra $(\mathbb{R}^{24}, +, \star)$ has exactly eight idempotents, by symbolic classification.[8] Writing $\varepsilon_1 = (e_0, 0, 0)$, $\varepsilon_2 = (0, e_0, 0)$, $\varepsilon_3 = (0, 0, e_0)$ and $\omega = \varepsilon_1 + \varepsilon_2 + \varepsilon_3$, they are

$$0, \qquad \varepsilon_1,\ \varepsilon_2,\ \varepsilon_3, \qquad \tfrac{1}{3}\omega, \qquad \varepsilon_i - \tfrac{1}{3}\omega \ \ (i = 1, 2, 3). \tag{5.1}$$

Note that every block of every one of them is a multiple of e_0.

Each nonzero idempotent spans a rational ray that meets Λ, and the least positive multiple that does is[9]

$$4\varepsilon_i \ (N = 16), \qquad 4\omega \ (N = 48), \qquad 6\big(\varepsilon_i - \tfrac{1}{3}\omega\big) \ (N = 24). \tag{5.2}$$

On the line spanned by a nonzero idempotent, the points of Λ are the integer multiples kw of the corresponding vector w in (5.2), of norm $16k^2$, $48k^2$ or $24k^2$; in particular, no real multiple of an idempotent lies on the minimal shell.

5.3 The automorphism group

$\mathrm{Aut}(\Lambda, +, \star)$ is finite, of order 36, with structure $C_6 \times S_3$. The S_3 is the block permutation of Definition 3.3, and the C_6 is generated by the following signed permutation of imaginary units, an automorphism of the twisted algebra $(\mathbb{O}, \cdot_\sigma)$ of Definition 3.2, applied in each block, with e_0 fixed:

$$e_1 \mapsto -e_5, \quad e_5 \mapsto e_6, \quad e_6 \mapsto e_1, \tag{5.3}$$

$$e_2 \mapsto -e_7, \quad e_7 \mapsto -e_3, \quad e_3 \mapsto e_2, \tag{5.4}$$

$$e_4 \mapsto -e_4. \tag{5.5}$$

The containment in the Conway group can already be settled at the level of the ambient algebra, before the lattice is imposed. As a real algebra, $\mathbb{R}^{24} = \mathbb{O}^3$ splits as $D \oplus T$, where $D = \{(a, a, a) : a \in$

[6] [repo] `python_project/src/verify_idempotents_min_shell.py`.
[7] [repo] `python_project/src/verify_pair_norm_histogram.py`.
[8] [repo] `python_project/src/verify_idempotent_classification.py`.
[9] [repo] `python_project/src/verify_idempotent_lattice_rescaling.py`.

$\mathbb{O}\}$ (of dimension 8) and $T = \{(p, q, r) \in \mathbb{O}^3 : p + q + r = 0\}$ (of dimension 16) are two-sided ideals with $D \star T = T \star D = 0$; every $\star$-automorphism preserves them and acts on each by an octonion automorphism, composed with a permutation of the blocks, so that $\mathrm{Aut}(\mathbb{R}^{24}, \star) = G_2 \times G_2 \times S_3$. An automorphism of the *real* octonions preserves the positive-definite octonion norm, and $D \perp T$; hence every $\star$-automorphism of $\mathbb{R}^{24}$ is orthogonal, and

$$\mathrm{Aut}(\Lambda, +, \star) \subsetneq \mathrm{Co}_0 . \tag{5.6}$$

The inclusion is strict because $-I_{24} \in \mathrm{Co}_0$ does not preserve $\star$.

The derivation, the complete enumerations, and an independent verification are recorded separately in a companion note.[10]

5.4 Square-zero elements

In the ambient algebra, $u = (x, y, z)$ satisfies $u \star u = 0$ if and only if x, y, z are purely imaginary, of equal norm, and sum to zero, again a symbolic classification. Three vectors of equal length summing to zero lie at 120° to one another; with their negatives they are the six vertices of a regular hexagon, and x, y, z are three alternating vertices, a *hexagonal triple* in $\Im(\mathbb{O}) = \mathbb{R}^7$. The solution set is a cone of dimension 12; it lies in T and meets D only in 0.[11]

The lattice does contain such elements. A vector $(x, y, z) \in \Lambda$ squares to zero exactly when x, y, z lie in $L\bar{s} \cap \Im(\mathbb{O})$, have equal norms, and sum to zero. The 126 vectors of norm 4 in $L\bar{s} \cap \Im(\mathbb{O})$ form an E_7 root system. Each of its 126 roots has exactly 32 roots at 120°; the third block is then forced, so the ordered hexagonal triples number $126 \times 32 = 4{,}032$, one square-zero vector of Λ each, all of norm 12. Every square-zero norm of Λ lies in $12\mathbb{Z}$ ($N(u) = 3N(x)$ with $N(x)$ even, and every norm of Λ lies in $4\mathbb{Z}$), and the 4,032 root triples are exactly the norm-12 stratum; higher strata are realized, for example, the blockwise sum of two blockwise orthogonal root triples is square-zero of norm 24.[12]

5.5 Other twists: closure and span of the image

The transposition σ is a special case of the permutations of the seven imaginary octonion basis elements, described by the symmetric group S_7. For a general such permutation π, the triple product $\star_\pi$ assembled blockwise from $x \cdot_\pi y := \pi^{-1}(\pi(x) \cdot \pi(y))$ may or may not close on Λ; for the involution σ, with $\sigma^{-1} = \sigma$, the factor $x \cdot_\sigma y$ is the product of Definition 3.2. When $\star_\pi$ closes, its *span index* is the index in Λ of the $\mathbb{Z}$-span of the products $u \star_\pi v$ ($u, v \in \Lambda$). By bilinearity, closure and span index are decided by the 576 basis-pair products of a $\mathbb{Z}$-basis of Λ. For $\star = \star_\sigma$ the span index is 2^{16}, and the span contains 2Λ.[13] For every cycle type of S_7 we find:[14]

[10] [repo] `paper/automorphism_group_2026-07-12.pdf`; [repo] `python_project/src/verify_aut_lambda_star.py` [repo] `gap_project/aut_lambda_star.g`.

[11] [repo] `python_project/src/verify_square_zero_classification.py`.

[12] [repo] `python_project/src/verify_square_zero_norm24_witness.py`.

[13] [repo] `python_project/src/verify_product_span_index.py` [repo] `python_project/src/verify_product_span_structure.py`.

[14] [repo] `python_project/src/verify_all_permutations_exact.py` [repo] `python_project/src/verify_twist_span_index_census.py`.

Cycle type of π	Permutations	Close	Rate	Span index
2-cycle	21	21	100%	2^{16}
3-cycle	70	35	50%	2^{10}
$(2,2)$	105	42	40%	2^{10}
4-cycle	210	0	0%	n/a
$(3,2)$	420	126	30%	2^{16}
5-cycle	504	252	50%	2^{10}
$(2,2,2)$	105	21	20%	2^{16}
$(3,3)$	280	112	40%	2^{10}
$(4,2)$	630	294	46.7%	2^{10}
6-cycle	840	210	25%	2^{16}
$(3,2,2)$	210	105	50%	2^{10}
$(4,3)$	420	168	40%	2^{16}
$(5,2)$	504	42	8.3%	2^{16}
7-cycle	720	336	46.7%	2^{10}
Total	5,039	1,764	35.0%	

The identity (the untwisted product) does not close. The 4-cycles are the only type failing without exception; the $(5,2)$ type has the lowest nonzero rate. The 3-cycles, 5-cycles, and $(3,2,2)$ types close at exactly half. For the 3-cycles the closing half admits a description: for each three-element subset of imaginary indices, one of the two cyclic orientations closes. No such description is known for the other cycle types, and none is known for the 0% of the 4-cycles. The span index depends only on the parity of π.

The transposition σ is also the reflection in the root $e_1 - e_2$ of L. With the reflection ρ in the root $r = \frac{1}{2}(e_0 + e_1 + \cdots + e_6 - e_7)$ of L in place of π, the triple product $\star_\rho$ closes on Λ with span index 1: its products span Λ.[15] The span index is invariant under ring isomorphism (for a unital order it would always be 1), so $\star_\sigma$, $\star_\pi$ for an even closing π, and $\star_\rho$ make Λ into three pairwise non-isomorphic orders.

6 Related work

Wilson (2009) [Wilson2009] characterizes Λ as $\{(x,y,z) \in L^3 : \text{conditions 1–3}\}$, gives an explicit listing of the 196,560 minimal vectors in the form $3 \times 240 \times (1 + 16 + 16 \times 16)$, and proves that Λ is even and self-dual in his halved-norm convention. The sublattice framework is the setting used in the present paper.

Dixon (2010) [Dixon2010] constructs Λ using the octonion XY-product[16]

$$A \circ_{XY} B \;=\; (AX)(Y^\dagger B), \tag{6.1}$$

where $X, Y \in \mathbb{O}$ are chosen so that $YX^\dagger$ is a unit octonion (the identity of the new multiplication). When $Y^\dagger = X^{-1}$ the XY-product specializes to the X-product $A \circ_X B = (AX)(X^{-1}B)$; Dixon

[15] [repo] `python_project/src/verify_twist_span_index_census.py`.

[16] The construction has an earlier precursor in Dixon [Dixon1995], where a closely related octonionic description of Λ is proposed. A 2005 preprint of Dixon's [Dixon2005] carried a formula whose third family of minimal vectors required a small correction, which Wilson [Wilson2009] supplied in a bracketed note while independently deriving the sublattice characterization. The situation is similar to the Kirmse–Bruck–Coxeter relationship recorded in Appendix B: an early, structurally correct proposal needing correction.

uses this X-product with $X = \ell_0 = \frac{1}{2}(1 + e_1 + \cdots + e_7)$ to obtain E_8, and general XY-products for higher-dimensional laminated lattices including Λ. We have independently reproduced this construction.[17]

Baez–Egan (2014) [Baez2014] study closure properties of integer sublattices of the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$. In parts 9–10 of the series, Baez considers the 27-dimensional lattice $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$, with integers on the diagonal and an off-diagonal Leech sublattice $L_L \cong \Lambda$ embedded as an octonionic triple. This lattice is *not* closed under the Jordan product $x \circ y = xy + yx$, but *is* closed under the doubled product $2(xy + yx)$. Egan further enumerates at least 17,280 distinct off-diagonal Leech embeddings $L_L \subset \mathbb{O}^3$ for which the associated 27-dimensional lattice is closed under this doubled product (see part 10 of [Baez2014] for Egan's computational counts). Baez leaves open the question whether any such lattice is closed under the undoubled Jordan product, and conjectures that the answer is negative. We have likewise independently reproduced the full Baez–Egan verification chain: the E_8 simple-roots data, the lemma $L_1 \cap L_2 = 2L_0$, the Leech-lattice theorem, the doubled-Jordan-product closure, the derivation of the lower bound 244,035,421 on the number of E_8^3-orbit Leech embeddings,[18] and Egan's count of at least 17,280 Jordan rings. This last factors as $17{,}280 = 270 \times 64$: the 270 is the number of sublattices $L_1 \subset L$ isometric to $\sqrt{2}\,L$ (part 11 of [Baez2014]), equivalently the number of 4-dimensional maximal totally isotropic subspaces of $L/2L \cong \mathbb{F}_2^8$ under the plus-type quadratic form induced by the E_8 Gram form; that mod-2 orthogonal-space picture is Tim Silverman's (part 12 of [Baez2014]). The 64 is the number of L_2 in this collection that meet a given L_1 only in $2L$, equivalently the number of 4-dimensional complementary isotropic subspaces; Section A.1 reads one such pair, $\sigma(L\bar{s})/2L$ and $\sigma(Ls)/2L$, as ideal and subalgebra under the product that $L/2L$ inherits from the octonion order L. All Baez–Egan claims hold under our reading.[19]

Comparison. The product $\star$ is bilinear on $\mathbb{R}^{24}$ itself. It does not require an enclosing 27-dimensional Jordan-algebraic ambient within which the Leech sits as a subspace (cf. [Baez2014]), and the cross-block S_3 extending the $\mathbb{Z}_3$ routing is supplied by the routing of Definition 3.3 rather than by the octonion product itself: Dixon's XY-product [Dixon1995, Dixon2010] is a useful device for describing each E_8 block, but it is not in itself $\mathbb{Z}_3$-symmetric across the three factors. Whether the Leech embedding $\sigma(\Lambda) \subset \mathbb{O}^3$ obtained by applying σ blockwise overlaps with one of the embeddings cataloged by Egan, or sits outside that family, is a question this paper does not address.

Anatomy of the Baez–Egan closure on Λ. The Baez–Egan closure is on the ambient 27-dimensional lattice $\widetilde{\Lambda} = \mathbb{Z}^3 \oplus \Lambda \subset \mathfrak{h}_3(\mathbb{O})$, not on Λ alone. For off-diagonal Hermitian matrices $X, Y \in \mathfrak{h}_3(\mathbb{O})$ representing $u = (x, y, z)$ and $v = (x', y', z')$ in $\Lambda \subset \mathbb{O}^3$, direct computation gives

$$2(XY + YX) \;=\; D \;+\; R, \tag{6.2}$$

where R is an off-diagonal residue lying in Λ and $D \in \mathbb{Z}^3$ is a diagonal piece whose entries are four times the block-pair Euclidean inner products of u and v:

$$D \;=\; 4\big(\langle x, x'\rangle + \langle y, y'\rangle,\ \langle x, x'\rangle + \langle z, z'\rangle,\ \langle y, y'\rangle + \langle z, z'\rangle\big), \tag{6.3}$$

so $\operatorname{tr} D = 8\langle u, v\rangle$ recovers the $\mathbb{R}^{24}$ inner product of u and v. Projecting onto the off-diagonal subspace defines a $\mathbb{Z}$-bilinear closed product $\varphi(u, v) := R$ on Λ: an order in the sense of Corollary 1.2,

[17] [repo] `python_project/src/leech_dixon.py` [repo] `python_project/src/e8_dixon.py`.
[18] An orbit-stabilizer bound, $\lceil 3!\,|W(E_8)|^3/|\mathrm{Co}_0|\rceil$; [repo] `python_project/src/verify_egan_lower_bound.py`.
[19] [repo] `python_project/src/egan_baez_count.py` [repo] `python_project/src/verify_phi_span_index.py`.

but obtained by projecting from a larger ambient with the inner-product shadow D discarded. For minimal Leech vectors $u \in \mathrm{Min}(\Lambda)$, squaring gives $\operatorname{tr} D = 8N(u) = 64$, so the projection is non-trivial on every minimal-vector input. The induced product φ is moreover commutative (the Jordan product is symmetric in its arguments), whereas $\star$ is not (Section 5); the two orders are therefore distinct bilinear products on $\mathbb{R}^{24}$ that both close on Λ, reaching that closure through different ambient algebras. The distinction is quantitative at the span level as well. Computed by the same Hermite-normal-form method as for $\star$ (Section 5.5),[20] the $\mathbb{Z}$-span of $\varphi(\Lambda, \Lambda)$ has index 2^{25} in Λ. Unlike for $\star$, that span does not contain 2Λ: its index exceeds $[\Lambda : 2\Lambda] = 2^{24}$.

Earlier precursors. Further back, several constructions foreshadow the octonion-triple and Jordan-algebra frameworks above. John H. Conway and Neil J. A. Sloane [ConwaySloane1982] give 23 constructions of Λ, one for each Niemeier lattice, including a Turyn-style construction from three copies of E_8. James Lepowsky and Arne Meurman [LepowskyMeurman1982] describe Λ as a sublattice of E_8^3 containing $2E_8^3$ and use this to generate the Conway group Co_0 via permutations of the three E_8 factors together with one extra automorphism. Noam D. Elkies and Benedict H. Gross [ElkiesGross1996] obtain Λ from an integer form of the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ as the orthogonal complement of a distinguished identity element, and show that the automorphism group of this integer form is generated by octonion reflections [*octave reflections*, in their terminology]. Each of these constructions is additive or geometric: none defines or tests a bilinear product on Λ.

Leonard E. Dickson (1923), Johannes Kirmse (1924), Kurt Mahler (1942), Richard H. Bruck and H. S. M. Coxeter (1946). For a full historical account, see Appendix B.

Recent experimental work in adjacent areas. Related questions in lattice theory and in nonassociative algebra are currently pursued through computer-assisted experiment, the method used here.

Gabriele Nebe (2022). In computational lattice theory, Kirschmer and Nebe classify binary Hermitian lattices over number fields by computer [KirschmerNebe2022]; Nebe's extremal even unimodular 72-dimensional lattice of minimum 8 is a tensor product of the Barnes lattice and the Leech lattice [Nebe2012].

Gerald Höhn and Geoffrey Mason (2023). Höhn and Mason determine the 290 fixed-point sublattices of the Leech lattice under Co_0 [HohnMason2016] and establish, by computer, that most vertex superalgebras of odd unimodular lattices of rank 24 carry an $N = 4$ superconformal structure [HohnMason2023]; Höhn and Seysen give a computer-assisted determination of the order of the Monster [HohnSeysen2025].

Vladimir Dotsenko (2025) and Murray R. Bremner (2016). In computational nonassociative algebra, Dotsenko investigates identities by computer experiment, with conjectures on nilpotence and the nil property [Dotsenko2025]; Bremner, Madariaga, and Peresi determine polynomial identities for the octonions using the representation theory of the symmetric group [BremnerMadariagaPeresi2016].

Daniele Corradetti (2026). A thematically adjacent recent construction is Corradetti's order on E_8 via the Okubo algebra [Corradetti2026]: a different non-classical bilinear structure on E_8, obtained by a 2-adic conductor / saturation of an Okubo-algebra overlattice. The two operations involved differ from the present mod-2 quotient (Appendix A, Section A.1): Corradetti rescales an order and 2-adically saturates a sublattice, while the present paper passes to a quotient $L \to L/2L$. Both place 2-primary structure at the center of an octonion-derived order on E_8.

[20] [repo] `python_project/src/verify_phi_span_index.py`.

7 Conclusion

The Leech lattice admits an order under a $\mathbb{Z}_3$-symmetric triple-octonion product on $\mathbb{R}^{24}$, realized in Wilson's sublattice representation of Λ. The symbolic proof (Section 4) reduces closure to four lemmas, of which the crux is Lemma 4.4: the sublattice $\sigma(Ls)$, unlike Ls itself, is closed under the standard octonion product.

The order is not maximal: it is properly contained in the $\star$-closed lattice L^3. The proof of Proposition 4.6 uses only Wilson's condition (1), that the blocks of u and v lie in L, so it shows $L^3 \star L^3 \subseteq L^3$; and L^3 is an even integral lattice containing Λ with index $[L^3 : \Lambda] = 2^{12}$. A second $\star$-closed overlattice is not contained in L^3. The idempotent $\varepsilon_1 = (e_0, 0, 0)$ of Section 5.2 acts on either side as the swap of blocks 2 and 3, $\varepsilon_1 \star v = v \star \varepsilon_1 = (x', z', y')$ for $v = (x', y', z')$, a swap that preserves Λ because Wilson's three conditions are symmetric in the blocks, and $\varepsilon_1 \star \varepsilon_1 = \varepsilon_1$; so $\Lambda + \mathbb{Z}\varepsilon_1$ is $\star$-closed by bilinearity. Since $4\varepsilon_1$ is the least positive multiple of ε_1 in Λ (Section 5.2), the index is $[\Lambda + \mathbb{Z}\varepsilon_1 : \Lambda] = 4$. This overlattice is not contained in L^3, because $e_0 \notin L$, and it is not integral: ε_1 has half-integer inner products with vectors of Λ.

Several questions remain immediately open:

- A structural reason for Lemma 4.4. The proof in Appendix A exhibits the 64 integer coefficients certifying $\sigma(Ls) \cdot \sigma(Ls) \subseteq \sigma(Ls)$, but does not explain *why* $\sigma(Ls)$ is closed under the standard octonion product while Ls is not. The mod-2 reduction in Section A.1 identifies what the appendix tables certify after reduction modulo $2L$: $\sigma(Ls)/2L$ is a subalgebra of $L/2L$, the reduction of the octonion order L modulo 2. One candidate shape for an integer-level reason is ruled out: $\sigma(Ls)$ is not an ideal of L on either side. Of the 64 basis-pair products, 32 of $L \cdot \sigma(Ls)$ and 43 of $\sigma(Ls) \cdot L$ lie outside $\sigma(Ls)$.[21] The companion sublattice $\sigma(L\bar{s})$ is by contrast a *two-sided* ideal of L, of which Lemma 4.3 is the left-handed half, and this ideal property has a structural reason: $\sigma(L\bar{s}) = 2O$ for a maximal order $O \supseteq L$ of the standard octonion product, so Lemma 4.3 follows from $O \cdot O \subseteq O$. Likewise $\sigma(Ls) = pO$ with $p = e_2 - e_4 + e_5 - e_6$, and the open question becomes why pO is closed.[22] (See the closing paragraphs of Section A.1 for a Lagrangian-polarization reframing of this open question as a research program.)
- Where $\mathrm{Aut}(\Lambda, +, \star) \cong C_6 \times S_3$ (Section 5.3) sits inside Co_0 up to conjugacy, and whether the order-6 octonion automorphism generating the C_6 admits an intrinsic description.
- Classification of the algebra $(\mathbb{R}^{24}, +, \star)$: it is nonassociative, nonalternative, nonflexible, and nonunital (as an order), placing it outside the classical families.
- A structural characterization of which permutation cycles close. Every cycle type of S_7 has been enumerated exactly (Section 5.5), but the pattern in the census table is described only for the 3-cycles. Open: a reason for the total failure of the 4-cycles and for the exact one-half rates beyond the 3-cycles; and linear maps of $\mathbb{R}^8$ that are not coordinate permutations.

8 Outlook

A more natural classification of $(\mathbb{R}^{24}, +, \star)$ may emerge from a *ternary* rather than binary viewpoint. The Leech lattice carries a prominent $\mathbb{Z}_3$ symmetry (the cyclic permutation of the three octonion

[21] [repo] `python_project/src/verify_sigma_Ls_ideal_exclusion.py`.
[22] [repo] `python_project/src/verify_sigma_sublattice_maximal_order.py`.

blocks), and $\star$ is assembled from nine bilinear building blocks routed by that $\mathbb{Z}_3$ action. Ternary composition algebras, as studied by Elduque [Elduque1996, Elduque2000, Elduque2023] and others [SmithVojtechovsky2022, KamiyaOkubo2015, Okubo1995], are typically nonassociative, noncommutative, nonalternative, and non-power-associative, the same negative classification $(\mathbb{R}^{24}, +, \star)$ satisfies. Yet they carry high internal symmetry, with order-3 automorphisms playing the central role: the Okubo algebra has automorphism group SU(3), and its product is built from an order-3 automorphism of the octonions. Jonathan I. Hall's monograph [Hall2019] establishes that Moufang loops and groups with triality are essentially the same object. Recasting $\star$ as a trace-like reduction of a ternary operation compatible with the $\mathbb{Z}_3$-block symmetry is a natural direction for further work.

Programs that model fundamental physics on normed-division-algebra, triality, S_3 symmetry, or symmetric-composition-algebra structures [GresnigtGourlayVarma2023, Koeplinger2023, FureyHughes2025, MarraniCorradettiZucconi2025] are one setting in which a non-classical bilinear product on $\mathbb{R}^{24}$ may be of interest.

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Declarations

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Data and code availability. All data and code are public: the complete research record, including every verification script cited in this paper, is available at the repositories of [repo].

Use of AI. AI assistance is a stated, material part of this work, including the drafting of the manuscript text; the methodology, the role of AI, and the human-designed verification and governance process are documented in Appendix C. The author directed the work prompt by prompt and reviewed the AI-assisted output; machine-obtained results were verified under the protocol documented there. The author takes full responsibility for the content, including the accuracy of the references.

A Explicit basis tables for Lemmas 4.2–4.4

We fix concrete $\mathbb{Z}$-bases for the lattices L, $\sigma(Ls)$, and $\sigma(L\bar{s})$, and tabulate the integer coefficients of the basis-by-basis products called for in the proofs of Lemmas 4.2, 4.3, and 4.4. Each table has 64 entries and is arranged so that the row labeled (i, j) gives the unique 8-tuple $(c_1, \ldots, c_8) \in \mathbb{Z}^8$ representing the corresponding product in the basis indicated. All entries are integers; closure under the standard octonion product follows by $\mathbb{Z}$-bilinearity. The same identities have been recomputed in two independent computer-algebra implementations as a cross-check (Appendix C).

Coordinates throughout are with respect to the basis $\{e_0, \ldots, e_7\}$ and the standard Fano triples (2.1).

The basis $\{L_k\}$. Concretely: $L_1 = s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$, $L_2 = e_0 + e_1$, $L_3 = e_0 - e_1$, and $L_{k+2} = e_0 + e_k$ for $k = 2, \ldots, 6$. The eight vectors are linearly independent over $\mathbb{Q}$ and span L over $\mathbb{Z}$.

$$L_1 = (\tfrac{-1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2},\ \tfrac{1}{2}) \tag{A.1}$$
$$L_2 = (1,\ 1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0) \tag{A.2}$$
$$L_3 = (1,\ -1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0) \tag{A.3}$$
$$L_4 = (1,\ 0,\ 1,\ 0,\ 0,\ 0,\ 0,\ 0) \tag{A.4}$$
$$L_5 = (1,\ 0,\ 0,\ 1,\ 0,\ 0,\ 0,\ 0) \tag{A.5}$$
$$L_6 = (1,\ 0,\ 0,\ 0,\ 1,\ 0,\ 0,\ 0) \tag{A.6}$$
$$L_7 = (1,\ 0,\ 0,\ 0,\ 0,\ 1,\ 0,\ 0) \tag{A.7}$$
$$L_8 = (1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 1,\ 0) \tag{A.8}$$

The basis $\{M_k\}$. Defined by $M_k := \sigma(L_k \cdot s)$ with $\sigma = (1\ 2)$ acting on coordinates by swapping e_1 and e_2. Spans $\sigma(Ls)$ over $\mathbb{Z}$.

$$M_1 = (\tfrac{-3}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2},\ \tfrac{-1}{2}) \tag{A.9}$$
$$M_2 = (-1,\ 0,\ 0,\ 0,\ 1,\ 0,\ 1,\ 1) \tag{A.10}$$
$$M_3 = (0,\ 1,\ 1,\ 1,\ 0,\ 1,\ 0,\ 0) \tag{A.11}$$
$$M_4 = (-1,\ 0,\ 1,\ 0,\ 0,\ 1,\ 0,\ 1) \tag{A.12}$$
$$M_5 = (-1,\ 1,\ 1,\ 0,\ 0,\ 0,\ 1,\ 0) \tag{A.13}$$
$$M_6 = (-1,\ 1,\ 0,\ 1,\ 0,\ 0,\ 0,\ 1) \tag{A.14}$$
$$M_7 = (-1,\ 0,\ 1,\ 1,\ 1,\ 0,\ 0,\ 0) \tag{A.15}$$
$$M_8 = (-1,\ 1,\ 0,\ 0,\ 1,\ 1,\ 0,\ 0) \tag{A.16}$$

The basis $\{N_k\}$. Defined by $N_k := \sigma(L_k \cdot \bar{s})$. Spans $\sigma(L\bar{s})$ over $\mathbb{Z}$.

$$N_1 = (2,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0) \tag{A.17}$$
$$N_2 = (0,\ 0,\ -1,\ 0,\ -1,\ 0,\ -1,\ -1) \tag{A.18}$$
$$N_3 = (-1,\ -1,\ 0,\ -1,\ 0,\ -1,\ 0,\ 0) \tag{A.19}$$
$$N_4 = (0,\ -1,\ -1,\ 0,\ 0,\ -1,\ 0,\ -1) \tag{A.20}$$
$$N_5 = (0,\ -1,\ -1,\ -1,\ 0,\ 0,\ -1,\ 0) \tag{A.21}$$
$$N_6 = (0,\ -1,\ 0,\ -1,\ -1,\ 0,\ 0,\ -1) \tag{A.22}$$
$$N_7 = (0,\ 0,\ -1,\ -1,\ -1,\ -1,\ 0,\ 0) \tag{A.23}$$
$$N_8 = (0,\ -1,\ 0,\ 0,\ -1,\ -1,\ -1,\ 0) \tag{A.24}$$

Lemma 4.2: $L \cdot L \subseteq L$**.**

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
1	1	-1	-1	-1	0	0	0	0	0
1	2	0	-2	-2	1	1	0	1	0
1	3	2	2	2	-1	-1	0	-1	0
1	4	0	-2	-2	0	1	1	0	1
1	5	2	1	2	-1	-1	0	0	-1
1	6	0	-1	-2	0	0	0	1	1
1	7	2	1	2	0	-1	-1	-1	0
1	8	2	2	2	-1	0	-1	-1	-1
2	1	2	1	2	-1	-1	0	-1	0
2	2	0	1	-1	0	0	0	0	0
2	3	0	1	1	0	0	0	0	0
2	4	0	0	-1	1	0	1	0	0
2	5	2	3	3	-1	0	-1	-1	-1
2	6	0	1	0	-1	0	1	0	0
2	7	0	0	-1	0	0	0	1	1
2	8	0	1	0	0	0	0	-1	1
3	1	0	-1	-2	1	1	0	1	0
3	2	0	1	1	0	0	0	0	0
3	3	0	-1	1	0	0	0	0	0
3	4	0	0	1	1	0	-1	0	0
3	5	-2	-3	-3	1	2	1	1	1
3	6	0	-1	0	1	0	1	0	0
3	7	0	0	1	0	0	0	1	-1
3	8	0	-1	0	0	0	0	1	1
4	1	2	2	2	-1	-1	-1	0	-1
4	2	0	1	0	1	0	-1	0	0
4	3	0	-1	0	1	0	1	0	0
4	4	0	-1	-1	2	0	0	0	0
4	5	0	-1	-1	1	1	0	1	0
4	6	0	0	-1	1	0	1	0	0
4	7	0	0	0	1	-1	0	1	0
4	8	2	2	3	0	-1	-1	-1	0

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
5	1	0	-1	-2	1	0	0	0	1
5	2	-2	-2	-4	1	2	1	1	1
5	3	2	2	4	-1	0	-1	-1	-1
5	4	0	0	0	1	1	0	-1	0
5	5	0	-1	-1	0	2	0	0	0
5	6	0	-1	-1	0	1	1	0	1
5	7	0	-1	-1	1	1	0	1	0
5	8	0	0	0	0	1	-1	0	1
6	1	2	1	2	0	0	-1	-1	-1
6	2	0	0	-1	1	0	1	0	0
6	3	0	0	1	-1	0	1	0	0
6	4	0	-1	0	1	0	1	0	0
6	5	0	0	0	0	1	1	0	-1
6	6	0	-1	-1	0	0	2	0	0
6	7	2	2	3	-1	-1	0	0	-1
6	8	0	-1	-1	0	1	1	0	1
7	1	0	-1	-2	0	1	1	0	0
7	2	0	1	0	0	0	0	1	-1
7	3	0	-1	0	0	0	0	1	1
7	4	0	-1	-1	1	1	0	1	0
7	5	0	0	0	-1	1	0	1	0
7	6	-2	-3	-4	1	1	2	2	1
7	7	0	-1	-1	0	0	0	2	0
7	8	0	0	-1	0	0	0	1	1
8	1	0	-2	-2	1	0	1	1	0
8	2	0	0	-1	0	0	0	1	1
8	3	0	0	1	0	0	0	-1	1
8	4	-2	-3	-4	2	1	1	1	2
8	5	0	-1	-1	0	1	1	0	1
8	6	0	0	0	0	-1	1	0	1
8	7	0	-1	0	0	0	0	1	1
8	8	0	-1	-1	0	0	0	0	2

Tab. 1: Coefficients c_k such that $L_i \cdot L_j = \sum_{k=1}^{8} c_k \, L_k$, for the bases defined above.

Lemma 4.3: $L \cdot \sigma(L\bar{s}) \subseteq \sigma(L\bar{s})$**.**

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
1	1	-1	-1	-1	0	0	0	0	0
1	2	0	-2	-2	0	1	1	0	1
1	3	2	2	2	0	-1	-1	0	-1
1	4	0	-1	-2	0	1	0	1	0
1	5	2	1	2	-1	-1	0	0	-1
1	6	0	-2	-2	1	0	0	1	1
1	7	2	2	2	-1	-1	-1	-1	0
1	8	2	1	2	0	0	-1	-1	-1
2	1	2	2	2	-1	-1	-1	0	-1
2	2	-1	0	-2	0	0	1	0	1
2	3	1	1	2	0	0	0	-1	0
2	4	0	0	-1	1	0	0	0	1
2	5	0	0	-1	0	1	1	0	0
2	6	1	1	1	0	-1	1	-1	0
2	7	1	2	2	-1	-1	0	0	0
2	8	1	1	1	-1	0	0	-1	1
3	1	0	-2	-2	1	1	1	0	1
3	2	1	2	2	0	0	-1	0	-1
3	3	-1	-1	0	0	0	0	1	0
3	4	0	0	1	1	0	0	0	-1
3	5	0	0	1	0	1	-1	0	0
3	6	-1	-1	-1	0	1	1	1	0
3	7	-1	-2	-2	1	1	0	2	0
3	8	-1	-1	-1	1	0	0	1	1
4	1	2	1	2	-1	-1	0	-1	0
4	2	0	0	-1	1	0	1	0	0
4	3	0	0	1	1	0	-1	0	0
4	4	0	-1	-1	2	0	0	0	0
4	5	2	2	3	0	0	-1	-1	-1
4	6	0	-1	0	1	0	1	0	0
4	7	0	-1	-1	1	0	0	1	1
4	8	0	0	0	1	0	0	-1	1

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
5	1	0	-1	-2	1	0	0	0	1
5	2	0	1	0	0	1	0	-1	0
5	3	0	-1	0	0	1	0	1	0
5	4	-2	-3	-4	2	2	1	1	1
5	5	0	-1	-1	0	2	0	0	0
5	6	0	-1	-1	0	1	1	0	1
5	7	0	0	-1	0	1	0	1	0
5	8	0	0	0	0	1	-1	0	1
6	1	2	1	2	0	0	-1	-1	-1
6	2	1	2	1	-1	0	0	0	-1
6	3	-1	-1	-1	1	0	1	1	0
6	4	0	1	0	1	0	0	0	-1
6	5	0	0	0	0	1	0	1	-1
6	6	-1	-2	-3	1	1	2	1	0
6	7	1	1	1	0	-1	0	1	-1
6	8	1	1	1	0	0	0	0	0
7	1	0	-1	-2	0	1	1	0	0
7	2	-1	-1	-2	0	1	1	1	1
7	3	1	0	2	0	-1	0	0	0
7	4	0	-1	-1	1	0	1	1	0
7	5	0	-1	0	0	1	1	0	0
7	6	-1	-2	-2	0	0	2	1	1
7	7	1	0	1	-1	0	0	1	0
7	8	-1	-3	-3	1	1	1	1	2
8	1	0	-2	-2	1	0	1	1	0
8	2	-1	-1	-3	1	1	0	1	1
8	3	1	2	3	-1	-1	-1	0	0
8	4	0	0	0	1	0	-1	1	0
8	5	0	-1	-1	0	1	0	1	1
8	6	-1	-1	-2	1	0	1	1	1
8	7	-1	-1	-2	0	0	0	2	1
8	8	1	1	1	0	-1	-1	0	1

Tab. 2: Coefficients c_k such that $L_i \cdot N_j = \sum_{k=1}^{8} c_k \, N_k$, for the bases defined above.

Lemma 4.4: $\sigma(Ls)\cdot\sigma(Ls)\subseteq\sigma(Ls)$**.**

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
1	1	-1	1	1	0	0	0	0	0
1	2	0	0	2	0	-1	-1	0	-1
1	3	-2	-2	-4	0	1	1	0	1
1	4	0	1	2	-2	-1	0	-1	0
1	5	-2	-1	-2	1	-1	0	0	1
1	6	0	2	2	-1	0	-2	-1	-1
1	7	-2	-2	-2	1	1	1	-1	0
1	8	-2	-1	-2	0	0	1	1	-1
2	1	-2	-3	-2	0	1	1	0	1
2	2	2	-1	1	0	0	0	0	0
2	3	2	3	3	0	-2	-2	0	-2
2	4	2	1	2	-1	0	-1	0	-2
2	5	4	4	7	-2	-3	-2	-1	-2
2	6	0	-2	-1	1	0	-1	2	0
2	7	2	1	2	0	-1	-2	-1	0
2	8	2	1	3	1	-1	-1	-1	-2
3	1	0	1	0	0	-1	-1	0	-1
3	2	-2	-3	-5	0	2	2	0	2
3	3	2	1	1	0	0	0	0	0
3	4	-2	-3	-6	1	2	1	2	2
3	5	0	-2	-3	0	1	2	1	0
3	6	0	-2	-3	1	0	1	0	2
3	7	2	3	2	-2	-1	0	-1	0
3	8	2	1	1	-1	1	-1	-1	0
4	1	-2	-1	-2	-1	1	0	1	0
4	2	0	-2	-1	-1	0	1	0	2
4	3	4	4	5	-1	-2	-1	-2	-2
4	4	2	1	1	-2	0	0	0	0
4	5	2	2	3	-2	-2	1	-1	-1
4	6	2	1	2	-1	-2	-1	0	0
4	7	4	5	7	-3	-2	-2	-3	-1
4	8	2	0	2	-1	0	0	-1	-1

i	j	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
5	1	0	1	2	-1	-2	0	0	-1
5	2	-2	-5	-6	2	1	2	1	2
5	3	2	3	2	0	-1	-2	-1	0
5	4	0	-1	-2	0	0	-1	1	1
5	5	2	1	1	0	-2	0	0	0
5	6	0	-1	-1	2	-1	-1	0	1
5	7	0	0	-1	0	-1	0	-1	2
5	8	2	2	2	0	-1	-1	-2	-1
6	1	-2	-2	-2	1	0	-1	1	1
6	2	2	1	2	-1	0	-1	-2	0
6	3	2	3	2	-1	0	-1	0	-2
6	4	0	0	-1	-1	2	-1	0	0
6	5	2	2	2	-2	-1	-1	0	-1
6	6	2	1	1	0	0	-2	0	0
6	7	2	4	3	-1	-1	-2	-2	-1
6	8	4	5	7	-2	-1	-3	-2	-3
7	1	0	2	2	-1	-1	-1	-2	0
7	2	0	-2	-1	0	1	2	-1	0
7	3	0	-2	-3	2	1	0	1	0
7	4	-2	-4	-6	1	2	2	1	1
7	5	2	1	2	0	-1	0	-1	-2
7	6	0	-3	-2	1	1	0	0	1
7	7	2	1	1	0	0	0	-2	0
7	8	0	-1	-1	1	2	0	-1	-1
8	1	0	1	2	0	0	-1	-1	-2
8	2	0	-2	-2	-1	1	1	1	0
8	3	0	0	-2	1	-1	1	1	0
8	4	0	1	-1	-1	0	0	1	-1
8	5	0	-1	-1	0	-1	1	2	-1
8	6	-2	-4	-6	2	1	1	2	1
8	7	2	2	2	-1	-2	0	-1	-1
8	8	2	1	1	0	0	0	0	-2

Tab. 3: Coefficients c_k such that $M_i\cdot M_j=\sum_{k=1}^{8}c_k\,M_k$, for the bases defined above.

A.1 The mod-2-quotient reading of Tables 2 and 3

The integer tables above are the explicit certificates of Lemmas 4.3 and 4.4; we now record the structural account of *what* they certify.

From $L\bar{s}\cap Ls=2L$ (Section 2.3) one has $2L\subseteq L\bar{s}$ and $2L\subseteq Ls$, and applying σ (which fixes $2L$),

$$2L\subseteq\sigma(L\bar{s}),\qquad 2L\subseteq\sigma(Ls). \tag{A.25}$$

Because $L\cdot L\subseteq L$ (Lemma 4.2) and the octonion product is $\mathbb{Z}$-bilinear, $(a+2c)\cdot b\equiv a\cdot b\pmod{2L}$ for $c\in L$, and likewise in the right factor. The product therefore descends to a well-defined $\mathbb{F}_2$-bilinear product $\bar{\mu}$ on the *mod-2 quotient*

$$\bar{L}:=L/2L\cong\mathbb{F}_2^8,\qquad \bar{\mu}\colon\bar{L}\times\bar{L}\to\bar{L}, \tag{A.26}$$

the reduction of the octonion order L modulo 2, an eight-dimensional $\mathbb{F}_2$-algebra.[23] The structure constants of $\bar{\mu}$ are exactly the entries of Table 1 reduced modulo 2.

Since $2L$ lies in both $\sigma(L\bar{s})$ and $\sigma(Ls)$, those sublattices are *full preimages*, under $L \to \bar{L}$, of $\mathbb{F}_2$-subspaces

$$V := \sigma(L\bar{s})/2L, \qquad W := \sigma(Ls)/2L, \tag{A.27}$$

and from $\sigma(L\bar{s}) + \sigma(Ls) = L$ and $\sigma(L\bar{s}) \cap \sigma(Ls) = 2L$ one obtains $\bar{L} = V \oplus W$ with $\dim V = \dim W = 4$. Reducing the closure statements modulo $2L$ then gives

$$\text{Lemma 4.3} \iff \bar{\mu}(\bar{L}, V) \subseteq V, \qquad \text{Lemma 4.4} \iff \bar{\mu}(W, W) \subseteq W. \tag{A.28}$$

That is: Table 2 certifies that V is a **left ideal** of $\bar{L}$; Table 3 certifies that W is a **subalgebra** of it. The integer tables and this mod-2-quotient description are complementary: the tables verify, the quotient explains. The descent and both equivalences were checked in exact arithmetic.[24]

A note on terminology: the construction is a *quotient*, the surjection $L \to L/2L$, and is named the *mod-2 quotient*. It is kept verbally distinct from a 2-adic scaling or saturation, the operations behind Corradetti's thematically adjacent construction [Corradetti2026], which is contrasted with the present one in Section 6.

Relation to prior work. The lattice-level picture of $\bar{L} \cong \mathbb{F}_2^8$ as an orthogonal space under the plus-type quadratic form induced by the E_8 Gram form, together with its 4-dimensional maximal totally isotropic subspaces (Lagrangians), is Silverman's (part 12 of [Baez2014]); see Section 6. The reading of the Leech construction in terms of the preimages in L of a *complementary pair* of such Lagrangians is David Calderbank's, in the discussion thread of part 9 [Calderbank2023]. Building on those posts and their surrounding discussion, the present construction equips $\bar{L}$ with the $\mathbb{F}_2$-bilinear product $\bar{\mu}$ inherited from the octonion product $L \cdot L \subseteq L$, and identifies the two Lagrangians of one specific complementary pair, $V = \sigma(L\bar{s})/2L$ and $W = \sigma(Ls)/2L$, as ideal and subalgebra of $(\bar{L}, \bar{\mu})$.

Polarization of $\bar{L}$ by σ. Direct verification yields two further structural facts about the decomposition $\bar{L} = V \oplus W$.[25] First, V is in fact a *two-sided* ideal of $\bar{L}$: the right-ideal direction $\bar{\mu}(V, \bar{L}) \subseteq V$ holds in addition to the left-ideal direction (Lemma 4.3). The right-ideal direction is not formally implied by the left-ideal direction, since $\bar{\mu}$ is noncommutative.[26]

Second, equip $\bar{L}$ with the natural plus-type quadratic form

$$q\colon \bar{L} \to \mathbb{F}_2, \qquad q(v + 2L) \;:=\; N(v)/2 \pmod 2, \tag{A.29}$$

well-defined because L is even ($N(v + 2u) \equiv N(v) \pmod 4$). For $L = D_8^+$ this is the standard plus-type form on $\mathbb{F}_2^8$ with Witt index 4. Direct verification shows that *both* V and W are totally isotropic under q (every element of either subspace has $q = 0$); each is therefore a maximal totally isotropic subspace, a *Lagrangian*. The decomposition $\bar{L} = V \oplus W$ is thus a *Witt decomposition of* $(\bar{L}, q)$ *into a complementary pair of Lagrangians*, a polarization of the orthogonal $\mathbb{F}_2$-space.

[23] It is not an octonion algebra in the composition-algebra sense: those are simple, whereas $\bar{L}$ has the proper two-sided ideal V exhibited below.

[24] [repo] `python_project/src/verify_mod2_quotient.py`.

[25] [repo] `python_project/src/verify_polarization_lagrangians.py`.

[26] Commutativity is a property of the algebra, not of a basis; what matters is *which order* is being reduced. Reducing the Gravesian order $\mathbb{Z}\langle e_0, \ldots, e_7\rangle$ gives a commutative $\mathbb{F}_2$-algebra, since $e_i \cdot e_j - e_j \cdot e_i = 2\, e_i \cdot e_j$ lies in twice that order. The order L also contains half-integer elements such as s, and for these the commutator $a \cdot b - b \cdot a$ need not lie in $2L$; direct computation confirms that $\bar{\mu}$ is noncommutative.

The structural reframing. In this picture, Lemmas 4.3 and 4.4 together express a single fact: *the construction of Section 3 produces a polarization of* $\bar{L}$, with one Lagrangian carrying the two-sided-ideal role (V) and the other carrying the subalgebra role (W). The open question recorded in Section 7, a structural reason for Lemma 4.4, accordingly opens into a research program: which polarizations of $\bar{L}$ arise from this construction, and which polarizations, beyond the one constructed here, lift to a closed bilinear product on Λ itself.

B Historical note: integral Cayley numbers, 1923–1946

The construction this paper builds on was developed across four primary papers between 1923 and 1946. Each was obtained in the original, and its key computational content re-derived in exact arithmetic in this project (for Mahler's covering-radius theorem, by exact-arithmetic sampling).[27]

Dickson (1923) [Dickson1923] gave the first construction: the Cayley–Dickson doubled product and an explicit 8-element basis of an order O_D, with all 64 basis products closing. His Theorem XV states three maximal orders; the true count is *seven*. The cause is in his derivation (p. 321): he assumes the orders contain the Hurwitz quaternions, and exactly three of the seven do.

Kirmse (1924) [Kirmse1924] took up the problem independently. His multiplication table is itself a correct octonion algebra, and he derived a correct *necessary* form for the maximal orders, without proving sufficiency; thirty E_8-type lattices realize the form. Among the eight of those lattices that are stable under multiplication by units (an unstated narrowing of the criterion, in our reconstruction), he exhibited the one, J_1 (p. 70), that is not an order. The remaining seven of his eight are the genuine maximal orders. Kirmse also develops an ideal-theoretic treatment of the integral octonions, a substantive companion contribution that is taken up again in Mahler (1942), and methodically takes algebra alternativity as a foundational constraint, anticipating the structural role alternativity plays in later integral-octonion arithmetic.

Mahler (1942) [Mahler1942] developed the arithmetic of a maximal order: for every Cayley number X there is an integral G with $N(X - G) \leq \frac{1}{2}$ (sharper than Dickson's 5/4), and every left or right ideal is principal, generated by an integral Cayley number of norm 1, 2, or 4.

Coxeter (1946) [Coxeter1946] identifies Kirmse's error with the same non-closure witness and records that the remedy is due to Bruck: "Bruck's domain J can be derived from J_1 by transposing two of the i's" (§4). This is a transposition of two imaginary basis units, a linear involution of $\mathbb{R}^8$; the $\binom{7}{2} = 21$ such transpositions yield the seven maximal orders in seven sets of three. His §§6–13 develop the $E_8/5_{21}$ geometry of the corrected order. A postscript (§14) records his after-the-fact awareness of Dickson and Mahler and identifies Dickson's undercount with exactly the cause above.

It is curious to find that the Coxeter–Bruck transposition would prove useful here: the same linear involution of $\mathbb{R}^8$, acting now on a different object. The Coxeter–Bruck transposition acts on a maximal-order candidate that sits asymmetrically in $\mathbb{R}^8$; the present σ acts on the coordinate-symmetric placement $L = D_8^+$ (where it is inert at the E_8 stratum) and does its work on the sublattice Ls of Wilson's Leech construction.

[27] [repo] `python_project/src/verify_dickson_1923.py` [repo] `python_project/src/verify_kirmse_1924.py` [repo] `python_project/src/verify_kirmse_1924_forensic.py` [repo] `python_project/src/verify_mahler_1942.py` [repo] `python_project/src/verify_coxeter_1946.py`. Verification notes accompany each, in `paper/reviews/`.

C Research methodology

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of directed prompts, governed by a written methodology (the "Manifesto") established at the outset.

The human researcher (the author) brought the initial hypothesis (motivated by prior work on Okubo algebras and autotopies of normed composition algebras), established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work: implementing octonion algebras, Wilson's Leech lattice construction, trial algebras, and closure tests, and contributed mathematical analysis. It also drafted the manuscript text and made every revision to it, at the author's prompt-by-prompt direction.

Structural proposals from a second AI system (the Hermes agent, Nous Research, running DeepSeek V4 Flash through OpenCode) prompted the Lagrangian framing of Section A.1; the AI agent (Claude Opus 4.7) verified and corrected them.[28]

Tests were written from first principles to independently verify every foundation. This independent verification is how we know that the properties stated in the literature are correctly implemented. Closure of the twisted product was first observed on over 12,000,000 random pairs of minimal vectors with zero failures, which prompted the search for a symbolic proof.

Symbolic proof and numeric validation. The proof of Theorem 1.1 given in Section 4 is symbolic: three of the four lemmas are established by exhibiting a finite list of integer coefficients (Appendix A); the fourth, $\sigma(L) = L$, is a parity argument. The conclusion follows by $\mathbb{Z}$-bilinearity. No floating-point or sampled arithmetic enters the proof. Independently, the same basis-by-basis identities that constitute the appendix tables were recomputed in two separate computer-algebra implementations as a cross-check: a Python implementation using `fractions.Fraction` exact rationals, and a GAP/LOOPS implementation using GAP's native exact rational arithmetic.[29] Both implementations verify the same basis-pair identities as the appendix tables. This redundancy is validation, not proof: it serves to detect transcription or implementation errors in the appendix tables, but the logical content of the proof rests on the tables themselves.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology was essential: understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach. The failure analysis of the untwisted product (Wilson's condition (3)

[28] The proposals verbatim, the verification, and the corrections: [repo] `paper/hermes_mod2_structure_supplement.pdf`; [repo] `python_project/src/verify_discovery1_W_subalgebra.py` [repo] `python_project/src/verify_discovery2_V_isotropic.py`.

[29] [repo] `python_project/src/symbolic_proof_checks.py` [repo] `gap_project/tests/test_lemmas.g`.

per Definition 2.1, exclusively Type 3×Type 3) directly pointed to the mechanism that the twist needed to correct.

Iterative convergence toward a robust result, including learning along the way what it is we are actually doing, is how study, research, and discovery have always proceeded. Mistakes were made on both sides: the AI agent occasionally departed from the agreed protocol or produced incorrect intermediate analyses, and the human author at times framed prompts under misconceptions or accepted intermediate results without sufficient scrutiny. In each case, the error was caught by the systematic cross-checking that the methodology requires (explicit verification, prompt-by-prompt review, and independent reimplementation in Python and GAP) and corrected forward. What is novel here is the ability to trace the entire path, beginning to end, through the complete list of prompts.

The full record is available at the repositories of [repo]: all prompts, code, evidence, and git history. The reader can form their own assessment of what this means for questions of discovery and attribution.

References

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[Baez2014] J. C. Baez (with contributions by G. Egan, T. Silverman and others), *Integral octonions* (parts 1–12), The n-Category Café, 2013–2016. The material discussed here appears in parts 9–12: Egan's Leech construction (part 9, 23 November 2014) and the Jordan-ring count (part 10, 1 December 2014), `https://golem.ph.utexas.edu/category/2014/11/integral_octonions_part_9.html`, `https://golem.ph.utexas.edu/category/2014/12/integral_octonions_part_10.html`; the 270 rescaled copies of the first E_8 shell (part 11, 12 December 2014) and Tim Silverman's account of the integral octonions modulo 2 (part 12, 31 January 2016), `http://math.ucr.edu/home/baez/octonions/integers/integers_11.html`, `http://math.ucr.edu/home/baez/octonions/integers/integers_12.html`.

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